

DISTRIBUTION OF THE PRODUCT OF DETERMINANTS OF RANDOM MATRICES CONNECTED WITH THE NONCENTRAL MATRIX VARIATE DIRICHLET DISTRIBUTION

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zonal polynomials.

Summary: In this paper the exact as well as the
asymptotic distributions of the product of
determinants of matrices which are jointly
distributed as noncentral matrix variate
Dirichlet have been derived. The complex case
and the linear case have also been studied

1. INTRODUCTION

Let $A_i: p \times p$, $i=1, \dots, \ell$ and $B: p \times p$ be independently distributed
according to $W_p(m_i, Z)$, $i=1, \dots, \ell$ and $W_p(n, Z, \Theta)$ respectively.
Here the matrix Θ is the noncentrality parameter. Let

$$L_i = E^{-\frac{1}{2}} A_i (E^{-\frac{1}{2}})',$$

$E^{\frac{1}{2}}$ being a lower triangular matrix such that $E^{\frac{1}{2}} (E^{\frac{1}{2}})' = E = \sum_{i=1}^{\ell} A_i + B$.

The joint density of $L_1, \dots, L_{\ell}$ is called the noncentral matrix
variate Dirichlet density. Let $A_i: p \times p$, $i=1, \dots, \ell$ and $B: p \times p$

be independently distributed according to $CW_p(m_i, \Sigma)$, $i=1, \dots, \ell$ and $CW_p(n, \Sigma, \Theta)$ respectively. Let

$$\tilde{L}_i = E^{-\frac{1}{2}} A_i (E^{-\frac{1}{2}})',$$

where $E^{\frac{1}{2}}$ is the lower triangular matrix with real diagonal elements such that $E = E^{\frac{1}{2}} (E^{\frac{1}{2}})' = \sum_{i=1}^{\ell} A_i + B$. The joint density of $\tilde{L}_1, \dots, \tilde{L}_{\ell}$ is the noncentral matrix variate Dirichlet density in the complex case.

The aim of this work is to study the distributions of $\prod_{i=1}^{\ell} |L_i|^m$ and $\prod_{i=1}^{\ell} |\tilde{L}_i|^m$. First in section 2, some results regarding the noncentral matrix variate Dirichlet density are given. In sections 3 and 4 the distributions of suitable functions of $\prod_{i=1}^{\ell} |L_i|^m$ and $\prod_{i=1}^{\ell} |\tilde{L}_i|^m$ are given. In section 5 these distributions are given in the linear case.

It may be noted that many test criteria proposed in multivariate statistical analysis for hypothesis testing depend on the distributions considered here (Anderson (1984), Gupta and Nagar (1986a)).

2. PRELIMINARIES

De Waal (1972) derived the following results in the real and complex cases.

Lemma 2.1. The joint density of the random matrices $L_1, \dots, L_{\ell}$ is given by

$$\begin{aligned}
 f(L_1, \dots, L_\ell) &= \frac{|2Z|^{-(m+n)/2} \text{etr}(-\Theta/2)}{\Gamma_P(n/2) \prod_{i=1}^{\ell} \Gamma_P(n_i/2)} \\
 &\cdot \prod_{i=1}^{\ell} |L_i|^{(m_i-p-1)/2} \left| I - \sum_{i=1}^{\ell} L_i \right|^{(n-p-1)/2} \\
 &\cdot \int_{E > 0} \text{etr}(-Z^{-1}E/2) |E|^{(m+n-p-1)/2} \\
 &\cdot {}_0F_1(n/2; \Theta Z^{-1} E^{1/2} (I - \sum_{i=1}^{\ell} L_i) (E^{1/2})') dE, \\
 &0 < L_i < I, \quad 0 < \sum_{i=1}^{\ell} L_i < I,
 \end{aligned}$$

where $m = \sum_{i=1}^{\ell} m_i$. Also,

$$(2.1) \quad \mathbb{E} \left[\prod_{i=1}^{\ell} |L_i|^{m_i} \right]^h = \frac{\Gamma_P[(m+n)/2] \prod_{i=1}^{\ell} \Gamma_P[m_i/2 + m_i h]}{\Gamma_P[(m+n)/2 + mh] \prod_{i=1}^{\ell} \Gamma_P(m_i/2)}$$

$$\cdot \text{etr}(-\Theta/2) {}_1F_1((m+n)/2; (m+n)/2 + mh; \Theta/2),$$

where ${}_0F_1$ and ${}_1F_1$ are the hypergeometric functions of matrix argument (Constantine (1963), James (1964)).

Lemma 2.2. The joint density of random matrices $\tilde{L}_1, \dots, \tilde{L}_\ell$ is given by

$$\begin{aligned}
 f(\tilde{L}_1, \dots, \tilde{L}_\ell) &= \frac{|Z|^{-(n+m)} \text{etr}(-\Theta)}{\tilde{\Gamma}_P(n) \prod_{i=1}^{\ell} \tilde{\Gamma}_P(m_i)} \\
 &\cdot \prod_{i=1}^{\ell} |\tilde{L}_i|^{m_i-p} \left| I - \sum_{i=1}^{\ell} \tilde{L}_i \right|^{(n-p)}
 \end{aligned}$$

$$\begin{aligned} & \int_{\mathbf{E}=\overline{\mathbf{E}}, >0} \text{etr}(-\mathbf{Z}^{-1}\mathbf{E}) |\mathbf{E}|^{(m+n-p)} \\ & \cdot {}_0\tilde{F}_1(n; \mathbf{E}\mathbf{Z}^{-1}\mathbf{E}^{\frac{1}{2}}(\mathbf{I} - \sum_{i=1}^{\ell} \mathbf{L}_i)(\mathbf{E}^{\frac{1}{2}})^{-1}) d\mathbf{E}, \\ & 0 < \tilde{\mathbf{L}}_i = (\overline{\mathbf{L}_i})^{-1} < \mathbf{I}, \quad 0 < \sum_{i=1}^{\ell} \tilde{\mathbf{L}}_i = (\sum_{i=1}^{\ell} \overline{\mathbf{L}_i})^{-1} < \mathbf{I}, \end{aligned}$$

where $m = \sum_{i=1}^{\ell} m_i$. Also

$$\begin{aligned} (2.2) \quad \mathbb{E} \left[\prod_{i=1}^{\ell} |\tilde{\mathbf{L}}_i|^{m_i} \right]^h &= \frac{\prod_{i=1}^{\ell} \tilde{\Gamma}_p(m_i+n) \tilde{\Gamma}_p(m_i+m_i h)}{\tilde{\Gamma}_p(m+n+mh) \prod_{i=1}^{\ell} \tilde{\Gamma}_p(m_i)} \\ &\cdot \text{etr}(-\mathbf{E}) {}_1\tilde{F}_1((m+n; m+n+mh; \mathbf{E})), \end{aligned}$$

where ${}_0\tilde{F}_1$ and ${}_1\tilde{F}_1$ are the hypergeometric functions of complex matrix argument (James (1964)).

3. DISTRIBUTIONS OF $\prod_{i=1}^{\ell} |\mathbf{L}_i|$ AND $\prod_{i=1}^{\ell} |\tilde{\mathbf{L}}_i|$

Let $W = -(m+n-u)q \ln \sum_{i=1}^{\ell} \left[|\mathbf{L}_i| / \left[\frac{m_i}{m} \right]^p \right]^{m_i/m}$, $0 < W < \infty$, where u and $q(>0)$ are constants to be specified later. Then the distribution of W is given by the following theorem.

Theorem 3.1. If the joint distribution of random matrices $\mathbf{L}_1, \dots, \mathbf{L}_{\ell}$ is noncentral matrix variate Dirichlet, then the c.d.f.

of $W = -(m+n-u)q \ln \left[\frac{\ell}{n} \left\{ |L_i| / \left[\frac{m_i}{m} \right]^p \right\}^{\frac{m_i}{m}} \right]$ is given by

$$\begin{aligned}
 (3.1) \quad P[W \leq x] &= 2^{p(\ell-1)/2} \pi^{(\ell-1)p(p+1)/4} \\
 &\cdot \frac{\ell}{n} \left[\frac{(m_i/m)^{m_i p/2 - p(p+1)/4}}{\Gamma_p(m_i/2)} \right] e^{tr(-\Theta/2)} \\
 &\cdot \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p((m+n)/2, \kappa)}{k!} C_{\kappa}(\Theta/2) \\
 &\cdot \sum_{r=0}^{\infty} D_r^{\kappa} \left[\frac{m+n-u}{2} \right]^{-r-\nu} \cdot G_{r+\nu}(2q, x), \quad 0 < x < \infty,
 \end{aligned}$$

where $\kappa = \{k_1 \geq \dots \geq k_p \geq 0\}$, $k_1 + \dots + k_p = k$, denotes a partition of k ; $\sum_{\kappa}$ denotes summation over all such partitions κ of k ; $C_{\kappa}(\cdot)$ is the corresponding zonal polynomial,

$$\Gamma_p(a, \kappa) = \pi^{p(p-1)/4} \prod_{j=1}^p \Gamma(a - \frac{j-1}{2} + k_j), \quad \Gamma_p(a) = \Gamma_p(a, 0),$$

$$\nu = np/2 + p(p+1)(\ell-1)/4 + k; \quad G_{\alpha}(2q, x) = \int_0^x g_{\alpha}(2q, u) du; \quad g_{\alpha}(2q, x)$$

is the gamma density $[(2q)^{\alpha} \Gamma(\alpha)]^{-1} x^{\alpha-1} \exp(-x/2q)$; D_r^{κ} is defined recursively by

$$(3.2) \quad D_r^{\kappa} = \frac{1}{r} \sum_{s=1}^r s Q_s^{\kappa} D_{r-s}^{\kappa}, \quad r=1, 2, \dots, D_0^{\kappa} = 1,$$

$$\begin{aligned}
 (3.3) \quad Q_s^{\kappa} &= \frac{(-1)^{s+1}}{s(s+1)} \sum_{j=1}^p \left[\sum_{i=1}^{\ell} \delta_i^{-s} B_{s+1} \left(-\frac{1}{2} \delta_i (n-u) - \frac{1}{2} (j-1) \right) \right. \\
 &\quad \left. - B_{s+1} \left(k_j - \frac{1}{2} (j-1-u) \right) \right] \quad s = 1, 2, \dots
 \end{aligned}$$

and $\delta_i = m_i/m$, $i=1, \dots, \ell$, $\delta_1 + \dots + \delta_{\ell} = 1$.

Proof. Using (2.1) and results on hypergeometric functions (James (1964)), we get the characteristic function of W as

$$(3.4) \quad \varphi_W(t) = E[\exp(\omega W t)] = e^{\text{etr}(-\Theta/2)} \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p((m+n)/2, \kappa)}{\ell \prod_{i=1}^{\ell} \Gamma_p(m_i/2)} \\ \cdot \frac{C_{\kappa}(\Theta/2)}{k!} A(t, \kappa)$$

where $\omega = \sqrt{-1}$ and

$$(3.5) \quad A(t, \kappa) = \pi^{(\ell-1)p(p-1)/4} \prod_{i=1}^{\ell} \delta_i^{\omega t q p(m+n-u)\delta_i} \\ \cdot \prod_{j=1}^p \left\{ \frac{\prod_{i=1}^{\ell} \Gamma \left[\delta_i \left[\frac{m+n-u}{2} \right] (1-2\omega t q) - \frac{\delta_i(n-u)-1+j}{2}}{\Gamma \left[\left[\frac{m+n-u}{2} \right] (1-2\omega t q) + k_j - \frac{j-1-u}{2}} \right]} \right\}.$$

Now applying Barnes' formula (Anderson, 1984, p. 312) to the logarithm of (3.5) and converting back, one gets

$$(3.6) \quad A(t, \kappa) = (2)^{p(\ell-1)/2} \pi^{(\ell-1)p(p+1)/4} \prod_{i=1}^{\ell} \delta_i^{(mp\delta_i/2 - p(p+1)/4)} \\ \cdot \left[\left[\left[\frac{m+n-u}{2} \right] (1-2\omega t q) \right]^{-\nu} \left[1 + \sum_{r=1}^{\infty} D_r^{\kappa} \left\{ \left[\frac{m+n-u}{2} \right] (1-2\omega t q) \right\}^{-r} \right] \right]$$

where ν and D_r^{κ} are given in the theorem. Substituting the expression for $A(t, \kappa)$ from (3.6) in (3.4), inverting the resulting expression of $\varphi_W(t)$ by recognizing that $(1 - 2\omega q t)^{-\alpha}$ is the characteristic function of the gamma density $g_{\alpha}(2q, x)$, we obtain the desired result.

Remark 3.1. When $\ell = 1$, theorem 3.1 above reduces to the theorem 3.1 of Gupta and Javier (1986).

Next the distribution of $\tilde{W} = -2(m+n-u)q \ln \frac{1}{m} \left[\prod_{i=1}^{\ell} |\tilde{L}_i| / \left[\frac{m_i}{m} \right]^{p_i} \right]^{m_i/m}$

where u and $q (>0)$ are constants, is given by the following theorem.

Theorem 3.2. If the joint density of complex random matrices $\tilde{L}_1, \dots, \tilde{L}_\ell$ is noncentral complex matrix variate Dirichlet, then the c.d.f. of

$\tilde{W} = -2(m+n-u)q \ln \left[\frac{1}{m} \left\{ \prod_{i=1}^{\ell} |\tilde{L}_i| / \left[\frac{m_i}{m} \right]^{p_i} \right\}^{m_i/m} \right]$ is given by

$$(3.7) \quad P[\tilde{W} \leq x] = 2^{p(\ell-1)} \pi^{p^2(\ell-1)/2} \frac{1}{m} \left[\frac{m_i p_i^2/2}{\tilde{r}_p(m_i)} \right] \text{etr}(-\Theta) \\ \cdot \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\hat{\Gamma}_p(m+n, \kappa)}{k!} \tilde{C}_{\kappa}(\Theta) \left[\sum_{r=0}^{\infty} J_r^{\kappa}(m+n-u)^{-r-\eta} G_{r+\eta}(2q, x) \right]$$

where $\kappa = \{k_1 \geq \dots \geq k_p \geq 0\}$, $k_1 + \dots + k_p = k$ denotes a partition of k ; $\sum_{\kappa}$ denotes summation over all such partitions κ

of k ; $\tilde{C}_{\kappa}(\cdot)$ is the corresponding zonal polynomial;

$$\tilde{r}_p(a, \kappa) = \pi^{p(p-1)/2} \prod_{j=1}^p \Gamma(a-j+1+k_j), \quad \tilde{r}_p(a) = \tilde{r}_p(a, 0),$$

$\eta = k + pn + p^2(\ell-1)/2$; J_r^{κ} is defined by

$$(3.8) \quad J_r^{\kappa} = \frac{1}{r} \sum_{s=1}^r s R_s^{\kappa} J_{r-s}^{\kappa}, \quad r = 1, 2, \dots, \quad J_0^{\kappa} = 1,$$

$$(3.9) \quad R_s^K = \frac{(-1)^{s+1}}{s(s+1)} \sum_{j=1}^p \left[\sum_{i=1}^{\ell} \delta_i^{-s} B_{s+1}^{(1-j-\delta_i(n-u))} B_{s+1}^{(k_j-j+1+u)} \right]$$

$$s = 1, 2, \dots,$$

and $\delta_i = m_i/m$, $i = 1, \dots, \ell$, $\delta_1 + \dots + \delta_{\ell} = 1$.

Proof. Using (2.2), results on the hypergeometric function, and following exactly similar steps as in the real case, we get the desired result.

Remark 3.2. When $\ell = 1$, theorem 3.2 above reduces to the theorem 3.1 of Gupta and Nagar (1986b).

4. ASYMPTOTIC DISTRIBUTIONS OF W AND $\tilde{W}$

In this section we derive the asymptotic expansions of the distributions of W and $\tilde{W}$.

Theorem 4.1. The asymptotic distribution of

$$W = -(m+n-u)q \ln \sum_{i=1}^{\ell} \left\{ |L_i| / \left[\frac{m_i}{m} \right] \right\}^{p \cdot m_i/m} \quad \text{is given by}$$

$$(4.1) \quad P[W \leq x] = \sum_{k=0}^{\infty} P(k, \text{tr}(\Theta/2)) G_{\nu}(2q, x) \\ + e^{\text{tr}(-\Theta/2)} \sum_{k=0}^{\infty} \sum_{\kappa} \omega_{1k} [G_{1+\nu}(2q, x) \\ - G_{\nu}(2q, x)] \cdot \frac{C_{\kappa}(\Theta/2)}{k!} + o((\rho m)^{-2})$$

$$\text{where } \nu = \left\{ k + \frac{np}{2} + \frac{p(p+1)(\ell-1)}{4} \right\},$$

$$P(k, tr \Theta/2) = e^{tr(-\Theta/2)} [tr \Theta/2]^k / k!,$$

$$\omega_{1k} = -\frac{1}{\rho} \left[\frac{1}{m} \sum_{j=1}^p k_j (k_j - j) + k(1 - \rho + \frac{n}{m}) \right] \quad \text{and}$$

$$\rho = 1 + \left\{ n(n-p-1) - \left[\sum_{i=1}^{\ell} (1/\delta_i) - 1 \right] (2p^2 + 3p - 1)/6 \right\} / m[2n + (p+1)(\ell-1)].$$

Proof. In the expression for the c.d.f. of W let

$$\begin{aligned} (4.2) \quad \Psi(\kappa) &= \frac{\Gamma_p((m+n)/2, \kappa)}{\ell \prod_{i=1}^{\ell} \Gamma_p(m_i/2)} \\ &= \pi^{-p(p-1)(\ell-1)/4} \frac{1}{\prod_{j=1}^p} \frac{\Gamma[(m+n-u)/2 + k_j - (j-1-u)/2]}{\prod_{i=1}^{\ell} \Gamma[\delta_i(m+n-u)/2 - (j-1+\delta_i(n-u))/2]} \\ &= (2)^{-p(\ell-1)/2} \pi^{-(\ell-1)p(p+1)/4} \frac{1}{\prod_{i=1}^{\ell} \delta_i} \frac{-(mp\delta_i - p(p+1)/2)/2}{\left[\frac{m+n-u}{2} \right]^{\nu} \left[1 + \sum_{r=1}^{\infty} D_r^{\kappa} \left[\frac{m+n-u}{2} \right]^{-r} \right]} \end{aligned}$$

where $D_r^{\kappa} = \frac{1}{r} \sum_{s=1}^r s Q_s^{\kappa} D_{r-s}^{\kappa}$, $r=1, 2, \dots$, $D_0^{\kappa}=1$, and

$$\begin{aligned} (4.3) \quad Q_s^{\kappa} &= \frac{(-1)^{s+1}}{s(s+1)} \sum_{j=1}^p \left[B_{s+1}(k_j - \frac{j-1-u}{2}) \right. \\ &\quad \left. - \sum_{i=1}^{\ell} \delta_i^{-s} B_{s+1}(\frac{1}{2}(-\delta_i(n-u) - j+1)) \right]. \end{aligned}$$

Substituting (4.2) in (3.1) and multiplying the two infinite series in the resulting expression for the c.d.f. of W , one obtains

$$(4.4) \quad P[W \leq x] = \sum_{k=0}^{\infty} P(k, \text{tr}(\Theta/2)) G_{\nu}(2q, x) + e^{\text{tr}(-\Theta/2)} \sum_{k=0}^{\infty} \sum_{\kappa} Q_{\ell}^{\kappa} \left[\frac{m+n-u}{2} \right]^{-1} \\ \cdot [G_{1+\nu}(2q, x) - G_{\nu}(2q, x)] + O((m+n-u)^{-2}).$$

Simplifying Q_{ℓ}^{κ} from (3.3) using the result on $B_2(\cdot)$, we get

$$(4.5) \quad Q_{\ell}^{\kappa} = \frac{1}{2} \left[- \sum_{j=1}^p k_j (k_j - j) + \left[\sum_{i=1}^{\ell} \delta_i^{-1} - 1 \right] \frac{p(2p^2+3p-1)}{24} \right. \\ \left. - \frac{pu}{4} (2n+(p+1)(\ell-1)) + \frac{np}{4} (n+\ell(p+1)) - ku \right].$$

Now choose u such that $\frac{np}{4} (n+\ell(p+1)) - \frac{pu}{4} (2n+(p+1)(\ell-1)) + \left[\sum_{i=1}^{\ell} \delta_i^{-1} - 1 \right] \cdot \frac{p(2p^2+3p-1)}{24} = 0$ and choose a number ρ such that $(m+n-u) = \rho m$. Obtaining the solutions and substituting them in (4.5) we get

$$Q_{\ell}^{\kappa} = -\frac{m}{2} \left[\frac{1}{m} \sum_{j=1}^p k_j (k_j - j) + k \left(1 - \rho + \frac{n}{m} \right) \right].$$

Substituting this last expression and $(m+n-u) = \rho m$ in (4.4) we get the desired result.

Remark 4.1. When $q=1$, theorem 4.1 given above reduces to De Waal's (1972) theorem 2 except the value of ρ therein which appears to be wrong. For $\ell=1$, our result reduces to those of Gupta and Javier (1986) theorem 4.1 (also see De Waal (1968)).

Theorem 4.2. The asymptotic distribution of

$$\tilde{W} = -2(m+n-u)q \ln \frac{1}{n} \left\{ \prod_{i=1}^{\ell} \left| \tilde{L}_i \right| / \left[\frac{m}{m} \right]^p \right\}^{m_1/m}$$
 is given by

$$P(\tilde{W} \leq x) = \sum_{k=0}^{\infty} P(k, \text{tr} \Theta) G_{\eta}(2q, x) + \text{etr}(-\Theta) \sum_{k=0}^{\infty} \sum_{\kappa} \omega_{1k} [G_{1+\eta}(2q, x) - G_{\eta}(2q, x)] \cdot \frac{\tilde{C}_{\kappa}(\Theta)}{k!} + O((\rho m)^{-2}), \quad 0 < x < \infty$$

where $\eta = k + np + p^2(\ell-1)/2$, $P(k, \text{tr} \Theta) = \text{etr}(-\Theta) [\text{tr} \Theta]^k / k!$,

$$\omega_{1k} = -\frac{1}{2\rho} \left[\frac{1}{m} \sum_{j=1}^p (k_j^2 - 2jk_j + k_j) + 2k(1 - \rho + \frac{n}{m}) \right], \quad \text{and}$$

$$\rho = 1 + \left\{ n(n-p) - \left[\sum_{i=1}^{\ell} \sigma_i^{-1} - 1 \right] (2p^2 - 1)/6 \right\} / m[2n + p(\ell - 1)].$$

Proof. Following exactly similar steps as in the proof of the theorem 4.1, one can easily prove the above result.

Remark 4.2. When $q=1$, theorem 4.2 given above reduces to De Waal's (1972) theorem 3. For $\ell=1$, our result reduces to those of Gupta and Nagar (1986b) theorem 4.1.

5. DISTRIBUTIONS IN THE LINEAR CASE

Consider the case when the rank of the noncentrality parameter matrix Θ is 1, and let λ be the only nonzero latent root. Now theorems 4.1 and 4.2 will be specialized for this situation.

Theorem 5.1. The asymptotic distribution of

$$W = -\rho m q \ln \frac{1}{m} \left\{ \left| L_1 \right| / \left[\frac{m}{m} \right] \right\}^{p \cdot m / m}, \quad \text{where } L_1, \dots, L_{\ell} \text{ follow a}$$

noncentral matrix variate Dirichlet distribution, in the linear case is given by

$$(5.1) \quad P[W \leq x] = \sum_{k=0}^{\infty} P(k, \lambda/2) G_{\nu}(2q, x)$$

$$+ \exp(-\lambda/2) \sum_{k=0}^{\infty} \omega_{1k} \frac{(\lambda/2)^k}{k!} [G_{1+\nu}(2q, x) - G_{\nu}(2q, x)] + O((\rho m)^{-2})$$

where $\omega_{1k} = -(1/\rho)[(1/m)(k^2 - k) + k(1 - \rho + n/m)]$, and $P(\cdot, \cdot)$, ρ and ν have been defined in theorem 4.1.

Proof. If λ is the only nonzero latent root of Θ , then the zonal polynomial $C_{\kappa}(\Theta/2)$ corresponding to a partition κ of k is given by

$$C_{\kappa}(\Theta/2) = \begin{cases} (\lambda/2)^k, & \text{if } \kappa = \{k\} \\ 0, & \text{otherwise.} \end{cases}$$

Applying this result to the right-hand side of (4.1) we get (5.1).

Theorem 5.2. The asymptotic distribution of

$$\tilde{W} = -2\rho m q \ln \frac{e'}{m} \left\{ |\tilde{L}_1| / \left[\frac{m}{m} \right]^P \right\}^{m_1/m}, \quad \text{where complex random matrices}$$

$\tilde{L}_1, \dots, \tilde{L}_e$ follow a noncentral complex matrix variate Dirichlet distribution, in the linear case is given by

$$P[\tilde{W} \leq x] = \sum_{k=0}^{\infty} P(k, \lambda) G_{\eta}(2q, x) + \exp(-\lambda) \sum_{k=0}^{\infty} \omega_{1k} \frac{\lambda^k}{k!} \cdot [G_{1+\eta}(2q, x) - G_{\eta}(2q, x)] + O((\rho m)^{-2}),$$

where $\omega_{1k} = -(1/2\rho)[(1/m)(k^2 - k) + 2k(1 - \rho + n/m)]$ and $P(\cdot, \cdot)$, ρ and η have been defined in theorem 4.2.

Proof. Similar to the proof of theorem 5.1.

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